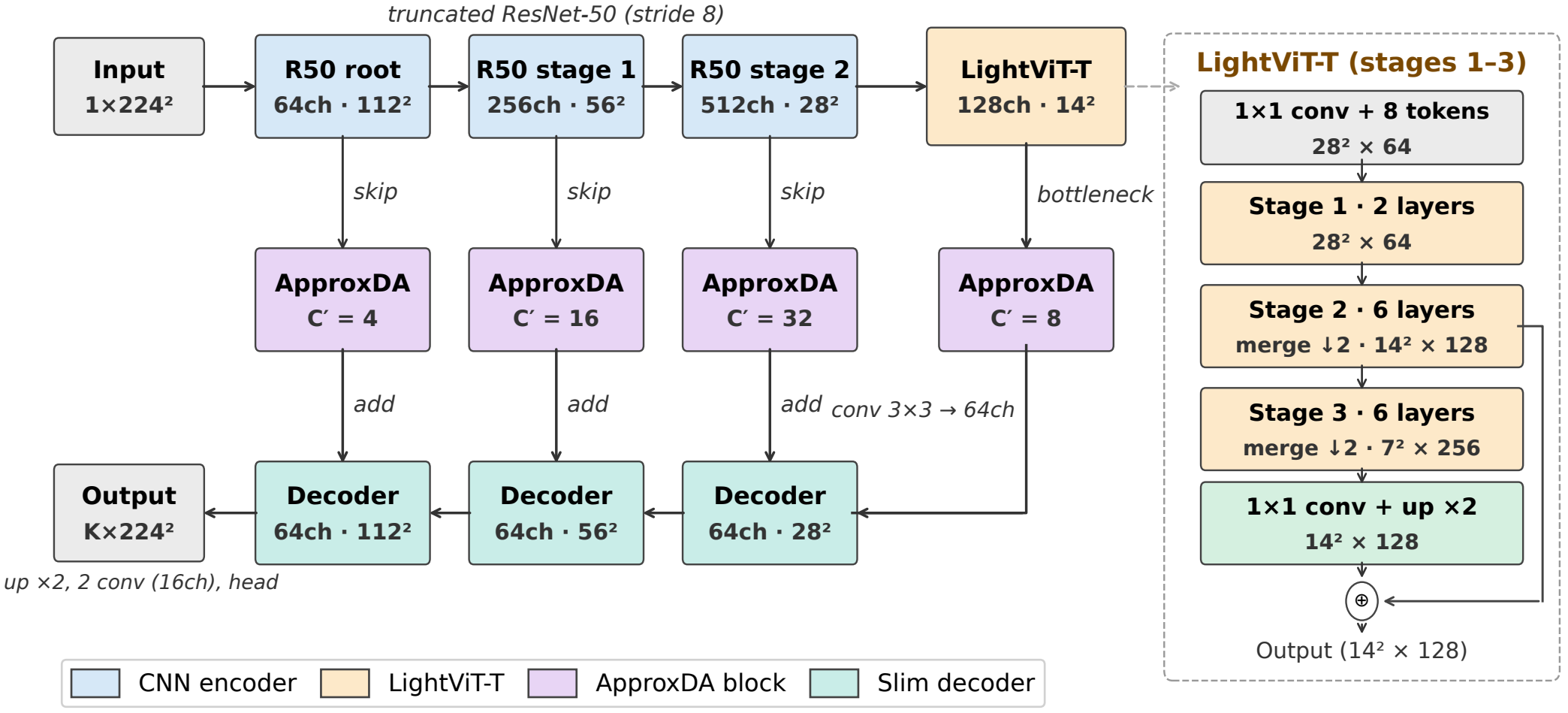
truncated ResNet-50 (stride 8)
Input
1×224²
R50 root
64ch · 112²
R50 stage 1
256ch · 56²
R50 stage 2
512ch · 28²
LightViT-T
128ch · 14²
skip
skip
skip
bottleneck
ApproxDA
C′ = 4
ApproxDA
C′ = 16
ApproxDA
C′ = 32
ApproxDA
C′ = 8
add
add
add
conv 3×3 → 64ch
Output
K×224²
Decoder
64ch · 112²
Decoder
64ch · 56²
Decoder
64ch · 28²
up ×2, 2 conv (16ch), head
LightViT-T (stages 1–3)
1×1 conv + 8 tokens
28² × 64
Stage 1 · 2 layers
28² × 64
Stage 2 · 6 layers
merge ↓2 · 14² × 128
Stage 3 · 6 layers
merge ↓2 · 7² × 256
1×1 conv + up ×2
14² × 128
⊕
Output (14² × 128)
CNN encoder
LightViT-T
ApproxDA block
Slim decoder